

Course Code: ME8104

Course Title: ADVANCED HEAT TRANSMISSION I

Semester: F2021

Instructor: Prof. Ziad Saghir

THERMAL PERFORMANCE OF TiO₂ NANO FLUID OVER WATER IN A HEATED PIPE

Title: Final Exam Project Report

Section Number:

Submission Date: 12/7/2021

Due Date: 12/7/2021

Name	Student Number (XXXX99999)	Signature
Godswill Ezeorah	501012886	G.E.

By signing above you attest that you have contributed to this submission and confirm that all work you have contributed to this submission is your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a "0" on the work, an "F" in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: www.ryerson.ca/senate/current/pol60.pdf.

Abstract

Heating and cooling have always been part of our life, including how foods are cooked, how we keep warm during the winter, etc. Similarly, for the home appliances like refrigerator and heating, ventilation and air conditioning systems (HVAC) some tubes or pipes are used to transporting, cooling or heat with fluids flowing through them (i.e., serving as a heat exchanger). This report presents a simple case of pipe with a fluid flowing through it to cool a constant parabolic heat flux Q . The simulation is done using COMSOL software to solve the problem using FEM for water and TiO_2 nanoparticle (of 1% of the base water volume). The comparison is also done for these two fluids at two different Reynolds numbers of 1000 and 2000 in the laminar flow regime. The numeric result is also compared with a simplified analytic solution, to validate the temperature behaviour along the heated pipe surface.

Table of Contents

Nomenclature	5
1. Introduction	6
2. Problem Description.....	6
3. Finite Element Method (FEM).....	7
3.1. Boundary Condition	8
3.2. Mesh Sensitivity	9
3.3. Simple Analytical Solution.....	10
4. Results and Discussion.....	11
4.1. Temperature Distribution and Flow Condition	11
4.2. Local Nusselt Number	14
4.3. Friction Factor Evaluation	16
4.4. Performance Evaluation for the Working Fluids.....	18
5. Conclusion.....	19
6. References	20
APPENDIX A	21

List of Tables

Table 1 Material Physical Properties	8
Table 2 Finite Element Mesh Information	9

List of Figures

Figure 1 Problem Description	7
Figure 2 Mesh Sensitivity	10
Figure 3 Analytical Solution for Temperature Distribution (with water as working fluid) ...	10
Figure 4 Temperature Distribution (with water as working Fluid)	11
Figure 5 Temperature Variation (with water as working Fluid)	12
Figure 6 Temperature Distribution (water vs. Nanofluid)	13
Figure 7 Velocity Flow Condition	14
Figure 8 Nusselt Number Variation (with water as the working fluid)	15
Figure 9 Nusselt Number comparison between Nano Fluid and Water	15
Figure 10 Friction Factor for Water (for radius, $r=0$)	16
Figure 11 Friction Factor for Water (for radius, $r=3$ mm)	17
Figure 12 1% TiO ₂ Nano Fluid vs. Water Friction Factor	17
Figure 13 Water vs. Nano Fluid Thermal Performance (w.r.t. Their Reynolds Number)	18
Figure 14 Water vs. Nano Fluid Thermal Performance (w.r.t $Re=1000$ of water fluid)	19

Nomenclature

TiO_2 – Titanium Dioxide

CuO – Copper Oxide

EG – Ethylene Glycol

FEM – Finite Element Method

Re – Reynolds Number

Nu – Nusselt Number

Vs. – Versus

1. Introduction

Cooling of heat-generating surfaces like the cup, electric vehicle batteries, can be challenging. Hence this paper aims to perform a simple heat exchanger analysis for the problem described below. But also to compare the cooling effect when the heat flux is increased in the heated surface. A review was done to consider CuO nanofluid as the working fluid for a comparative study against water. However, the thermal conductivity and dynamic viscosity value given by [2], might result in an underperformance when compared to water as the working fluid. It is believed this is caused by the base fluid for the CuO nanofluid, which was 70% water plus 30% ethylene glycol (EG). The high dynamic viscosity is caused by both the CuO particles (because of the high density of Copper (Cu)), and the low thermal conductivity is solely due to EG. Hence, until further research is done to obtain the physical properties for CuO nanofluid in 100% water base fluid, the 1% of TiO_2 nanofluid in water base from [1] is considered for the analysis done in this paper.

2. Problem Description

This paper aims to observe which of the working fluid, will have better thermal efficiency for an observing component of a heat exchanger. This component consists of an aluminum circular pipe with an inner radius (r_{in}) of 3.4 mm and a layer thickness of 0.6mm. The pipe is divided into three parts along its 85mm length, with $L_1 = 27.2$, $L_2 = 37.4$ and $L_3 = 20.4$. Part 1 and part 3 are thermally insulated at the outer surface of the pipe. While a parabolic heat flux q , is constantly applied on the outer surface of part 2. This heat flux is governed by,

$$q = q_0 \left(1 - \frac{4z^2}{L_2^2} \right) \quad (1.1)$$

Where, q_0 is a constant with the independent values $50,000 \text{ W/m}^2$ and $75,000 \text{ W/m}^2$, which are to be separately studied for the temperature distribution on just the part 2 surfaces. Steady-state condition is assumed for this problem, with the constant temperature at the inlet of the pipe as, $T_{in} = 18^\circ \text{ C}$. The fluid flow is considered for a laminar flow of Reynolds number, $Re = 1000$, and then compared with an increased flow velocity, i.e., for $Re = 2000$. With this the flow rate, Q can be calculated using,

$$Q = \frac{\pi r_{in} \mu Re}{2 \rho_f} \quad (1.2)$$

The above statement is described in the figure below for a clearer understanding of the problem.

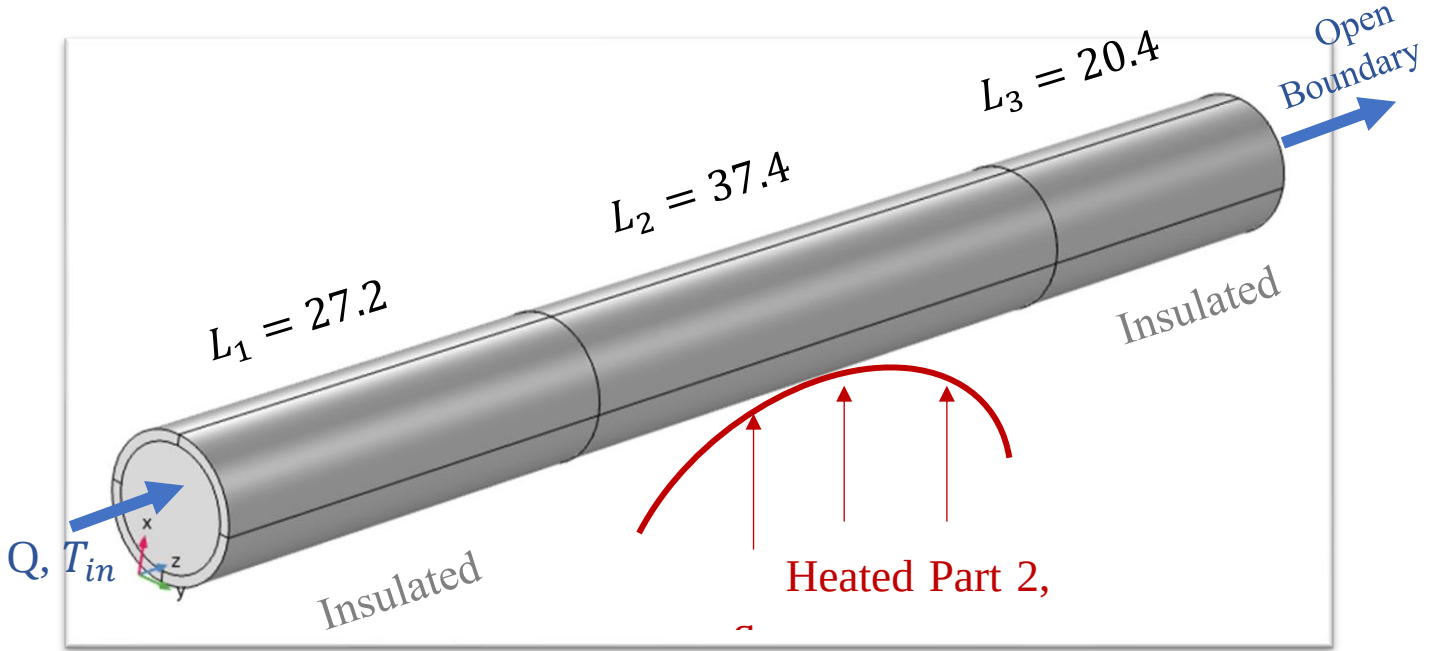


Figure 1 Problem Description

3. Finite Element Method (FEM)

To solve the above-presented problem, a numeric finite element formulation is developed from the Navier stokes, continuity and energy equation given below. This analysis is carried out using the COMSOL software (version 5.6, please see [4] for more details). The solver first state by calculating the velocity for an incompressible flow, using the continuity equation as,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

And then solving the pressure using the steady-state Navier stokes momentum conservation equation in the cartesian coordinate system, which is given as,

In x-direction,

$$\rho_f \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = -\frac{\partial p}{\partial x} + \mu_f \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

In y-direction,

$$\rho_f \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) = -\frac{\partial p}{\partial y} + \mu_f \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) + \rho_f g$$

In z-direction,

$$\rho_f \left(u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu_f \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right)$$

And the temperature distribution T , inside the fluid, is gotten us the energy conservation equation,

$$\rho_f C_{pf} \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right) = K_f \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right)$$

Where, ρ_f , C_{pf} , K_f and μ_f are the density, heat capacity, thermal conductivity and dynamic viscosity of the working fluid respectively. And u , v and w are the velocity components of the fluid flow in the x , y and z directions respectively.

3.1. Boundary Condition

At the inlet, the temperature is set as T_{in} and the flow rate Q , is solved directly by defining equation (1.2) in the COMSOL parameter definition menu, this will solve for when the Reynolds number is 1000 and 2000. The outflow region is set to an open boundary with zero normal share force. And at the wall, the insulated parts (1 and 3) automatically have zero heat flux, whereas the Neumann boundary condition is applied at part 2, in the form of the steady-state Fourier heat conduction equation for an anisotropic aluminum medium as,

$$\frac{dT}{dx} = -\frac{q}{k_s} \quad (3.1)$$

This is derived by analytically solving a 1D energy equation of, $\frac{\partial^2 T}{\partial x^2} = -\frac{q}{k_s}$

So that for solving the parabolic heat flux q in equation (1.1), the position variable z , the origin is shifted to the center of part 2 as,

$$z = z_0 - \left(L_1 + \frac{L_2}{2} \right)$$

The physical property of the materials used for this project are provided in the table below (TiO_2 nanofluid property is taken from [1])

Table 1 Material Physical Properties

Materials	Density, ρ (kg/m^3)	Dynamic Viscosity, μ ($Pa.s$)	Thermal Conductivity, K ($W/m.K$)	Heat Capacity, C_p ($J/kg.K$)
Water (@about 20° C)	998.03	0.001002	0.598	4184
TiO_2 Nanofluid	1030	0.001019	0.835	4040
Aluminum	2710	NA	237	910

3.2. Mesh Sensitivity

To determine the mesh sensitivity of the finite element analysis of COMSOL software, the average Nusselt number is calculated at the inner wall surface of part 2. The formula for calculating the Nusselt number is given as,

$$Nu = \frac{2*r_{in}*h}{k_f} \quad (3.2)$$

Where the heat convection coefficient is given as,

$$h = \frac{q}{T - T_{in}}$$

For the various levels of mesh available in the software, the following mesh information was obtained,

Table 2 Finite Element Mesh Information

Normal	249684 domain elements, 29586 boundary elements, and 1472 edge elements
Coarse	131957 domain elements, 19400 boundary elements, and 1196 edge elements
Coarser	51800 domain elements, 9746 boundary elements, and 816 edge elements
Extra Coarse	25698 domain elements, 5752 boundary elements, and 644 edge elements

And the resulting solution for the different levels of mesh had a maximum of about 4.5% deviation from each other. This means any of the mesh can be selected and will yield accurate numeric results.

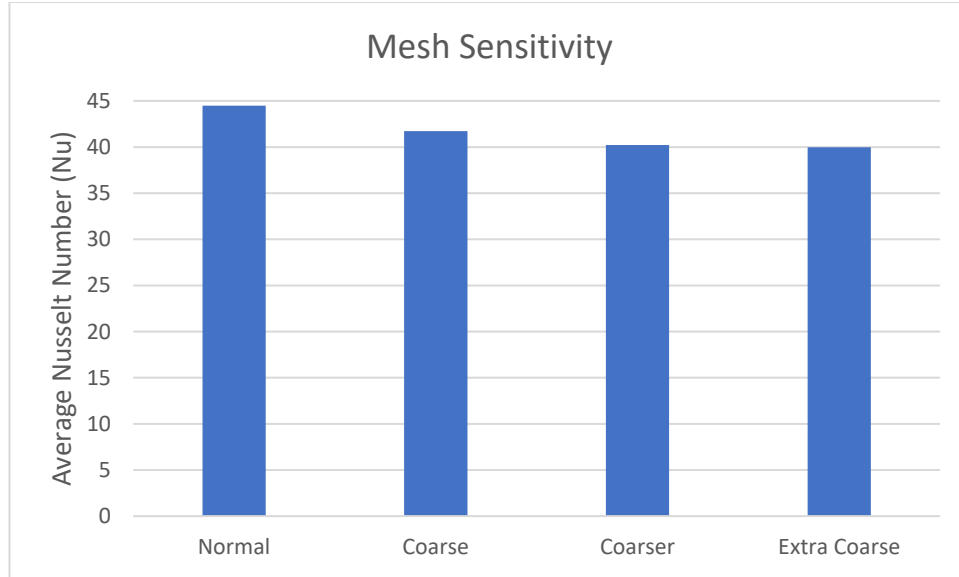


Figure 2 Mesh Sensitivity

For this paper, the Coarser mesh level has been selected for efficient compute time and result.

3.3. Simple Analytical Solution

In order to be sure that the numeric solution behaves as it should, a simple conduction heat transfer problem is analyzed just for the pipe wall (of thickness $t = 0.0006 \text{ m}$), equation (3.1) is used for this analysis with the MATLAB code provided in Appendix A. Below is presented the temperature distribution at the external wall using the analytical solution.

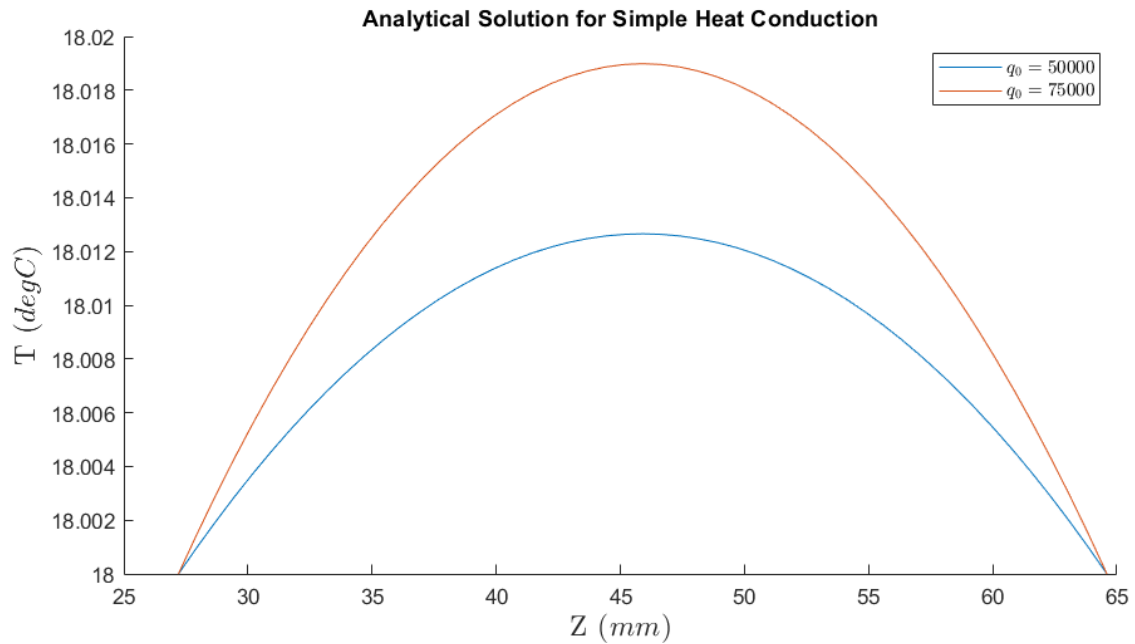


Figure 3 Analytical Solution for Temperature Distribution (with water as working fluid)

The above plot agrees with the parabolic behaviour of the inward heat flux q . And thus the numerical result should have similar temperature distribution at the external wall surface for the heated part 2, but with a variable inside temperature due to the laminar fluid flow within the pipe and its cooling effect.

4. Results and Discussion

The thermal enhancement is numerically evaluated and the result comparisons are discussed for two different working fluids (water and 1% TiO_2 in water base, nanofluid). And the effect of changing the fluid flow rate through the Reynolds number is also compared in the laminar flow regime. Also, it is important to observe the thermal behaviour when the heat flux at the heated part2 is increased.

4.1. Temperature Distribution and Flow Condition

Focusing on the water as the working fluid, the figure below shows the temperature distribution at the external wall surface of the heated part 2. And the temperature variation at the internal wall surface of the pipe is also shown for this heated part. Also, to compare the effect of Reynolds number on the heat enhancement of the heat exchanger for different heat flux conditions at the heated part is shown below:

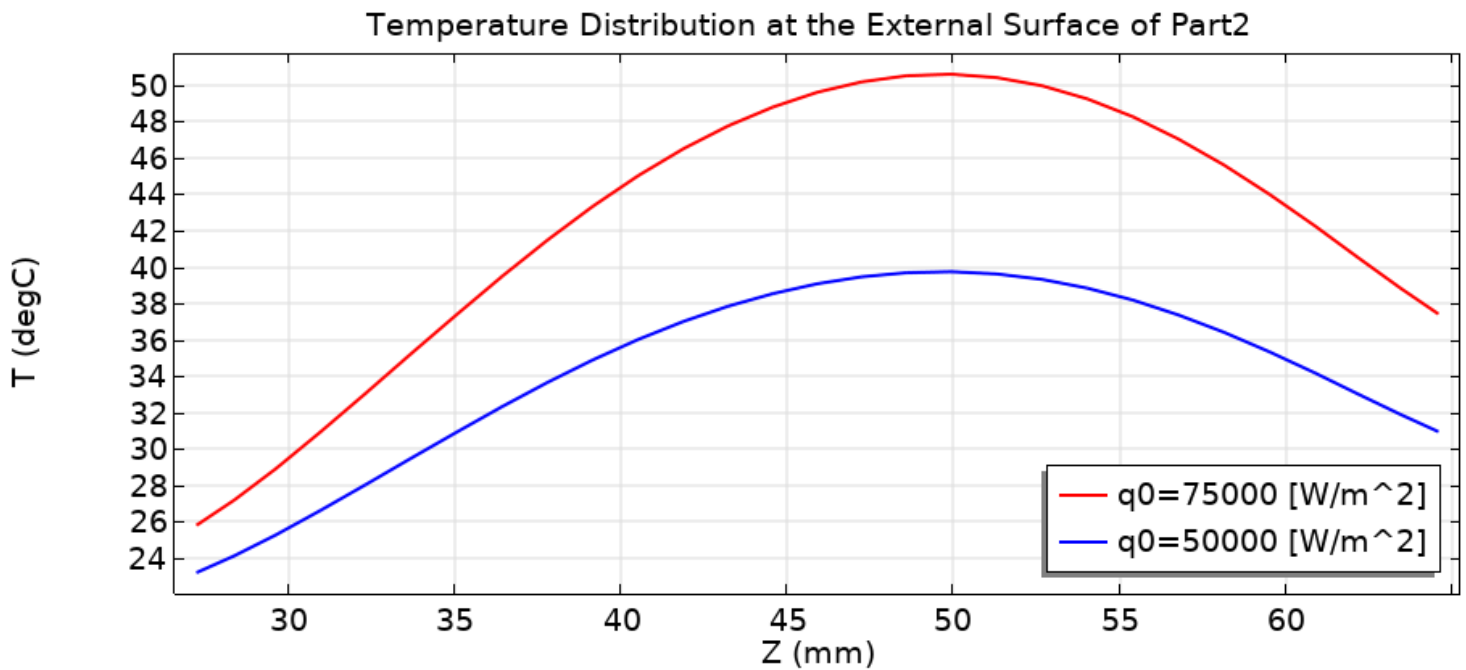


Figure 4 Temperature Distribution (with water as working Fluid)

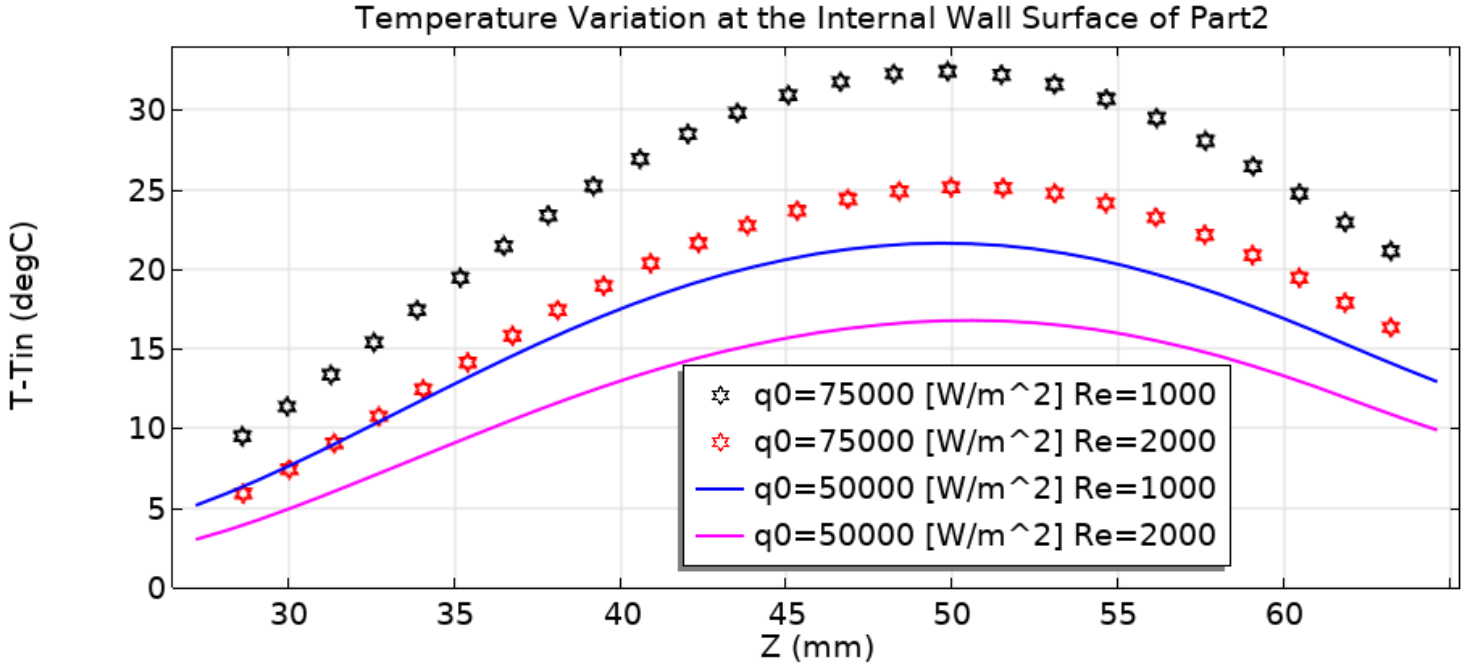


Figure 5 Temperature Variation (with water as working Fluid)

As observed the temperature distribution profile along the heated pipe length is similar to the analytical solution profile (in [Figure 3](#)) due to the parabolic nature of the heat flux. But for FEM analysis the temperature in the inside wall surface varies due to the fluid cooling effect unlike the constant temperature assumed in the simple analytical solution. This means that for the two independent heat fluxes the temperature variation is different and is non-linear at the internal wall surface due to convective heat transfer. And this translates to the external wall surface as well (i.e., they will have a similar heat distribution profile) at the heated part 2.

Also, the temperature differences between the internal wall surface and the maintained inlet fluid temperature (T_{in}), at the heated part, shows that the temperature variation decreases as the Reynolds number increases and as the heat flux input reduces.

The below figure shows the comparison between water and 1% TiO_2 Nanofluid as working fluids in terms of the temperature distribution at the internal wall surface of the heated part.

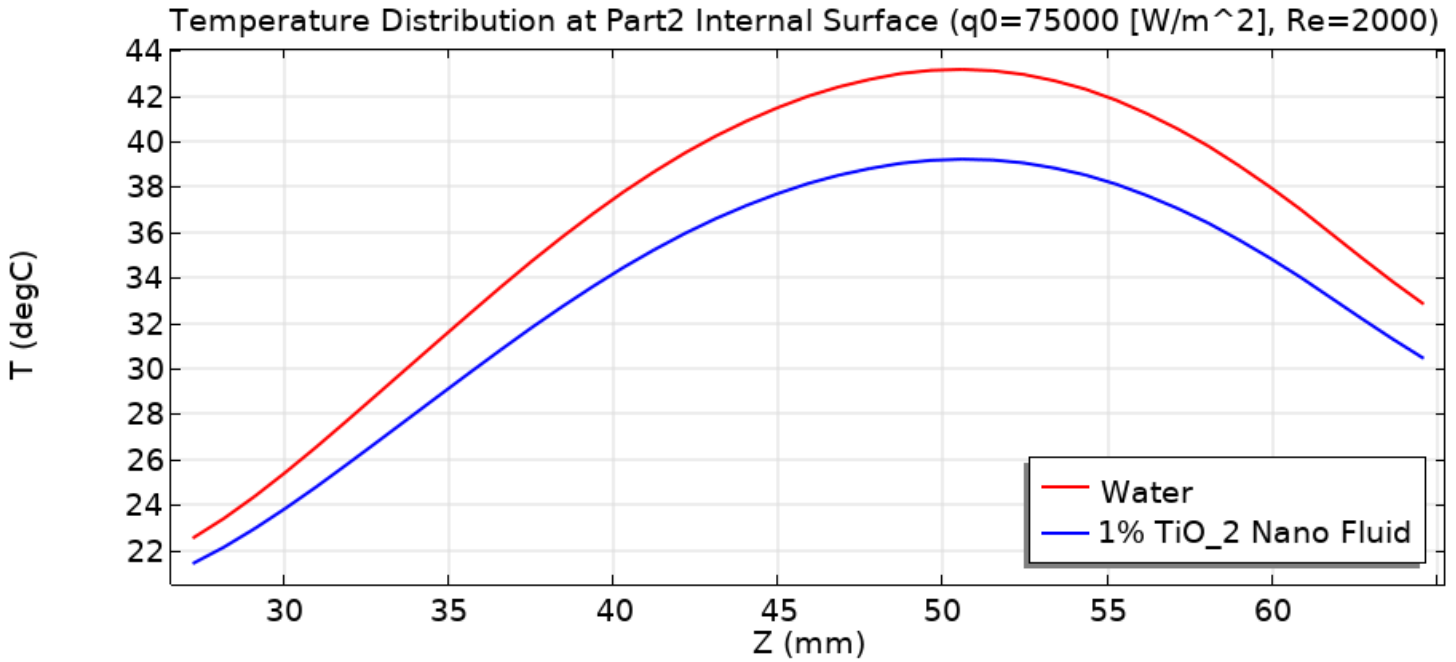


Figure 6 Temperature Distribution (water vs. Nanofluid)

It is observed that the nanofluid has better heat enhancement than water in cooling the internal surface. This is due to the almost identical viscosity of both fluids but a higher thermal conductivity with the nanofluid due to the 1% TiO_2 particles present in the water-based liquid. The effect of using this nanofluid will further be evaluated for its thermal efficiency when compared to water because it is not enough to have improved thermal enhancement but it is also important to evaluate the overall efficiency of the working fluid,

The flow condition of the fluid (water) is shown below for Reynolds number of 1000 and heat flux of $50,000 \text{ W/m}^2$:

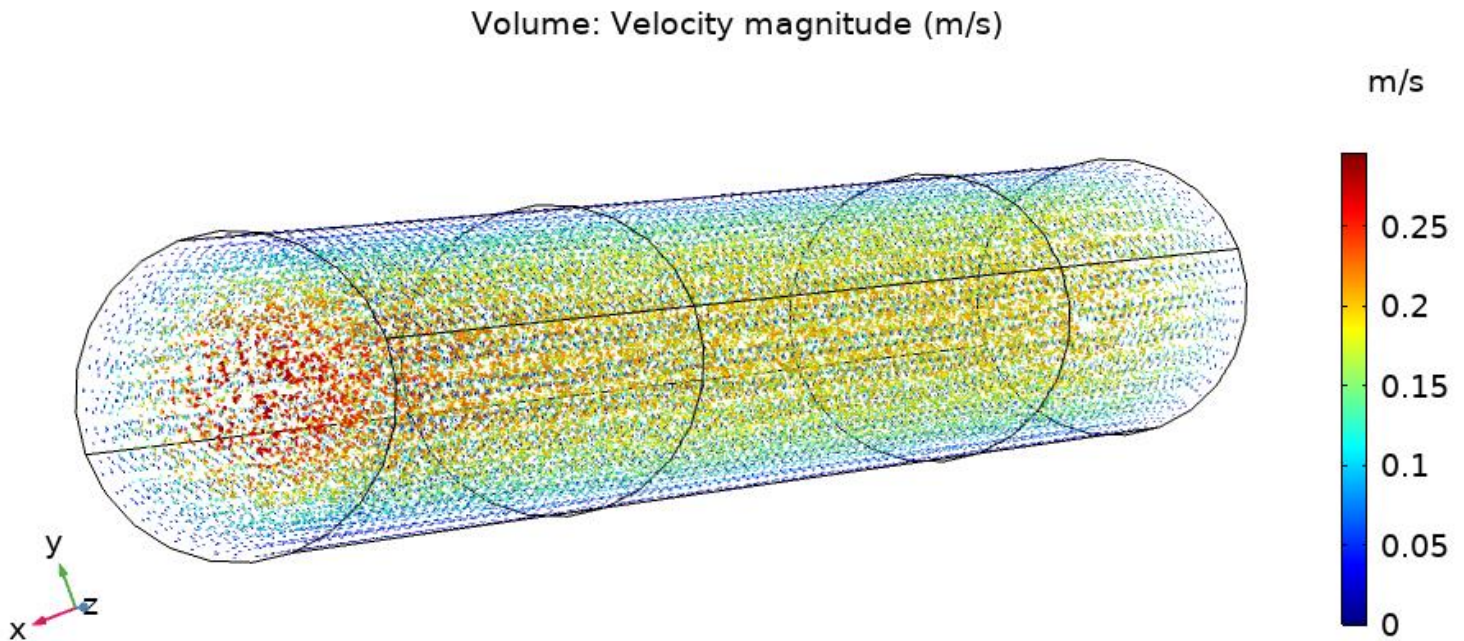


Figure 7 Velocity Flow Condition

The velocity is observed to be at its maximum at the center of the inlet, and then gradually slows down due to the no-slip condition, caused by the fluid viscosity. So that at the internal wall surface the flow condition of the velocity is approximately zero as shown above in the laminar flow regime.

4.2. Local Nusselt Number

The effect of Reynolds number variation and heat flux variation on the local Nusselt number for the heated part 2 was determined using equation (3.2). The resulting plot shows that increasing the heat flux has an almost negligible effect on the Nusselt number. But increasing the Reynolds number is observed to increase the Nusselt number, which in turn is responsible for the heat enhancement of the working fluid (i.e., water in this case). This is primarily achieved because of the increased fluid flow rate or velocity, which leads to enhanced cooling of the internal surface of the heated part 2. In translation, this property also shows the directly proportional relationship of the effect of increasing viscosity to the Nusselt number (i.e., increased fluid viscosity will reduce the fluid velocity and consequently reduce the heat enhancement of the working fluid, as shown by equation (1.2)).

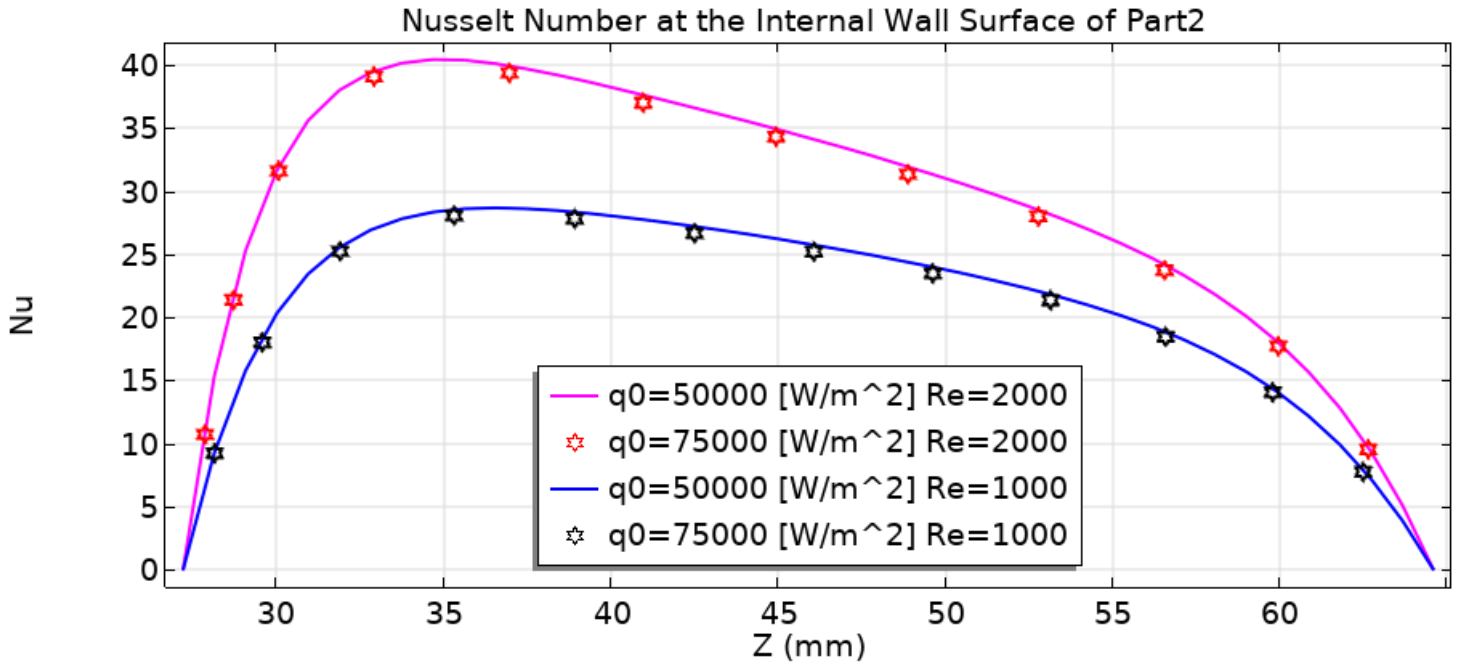


Figure 8 Nusselt Number Variation (with water as the working fluid)

The Nusselt number comparison between both working fluids are shown below:

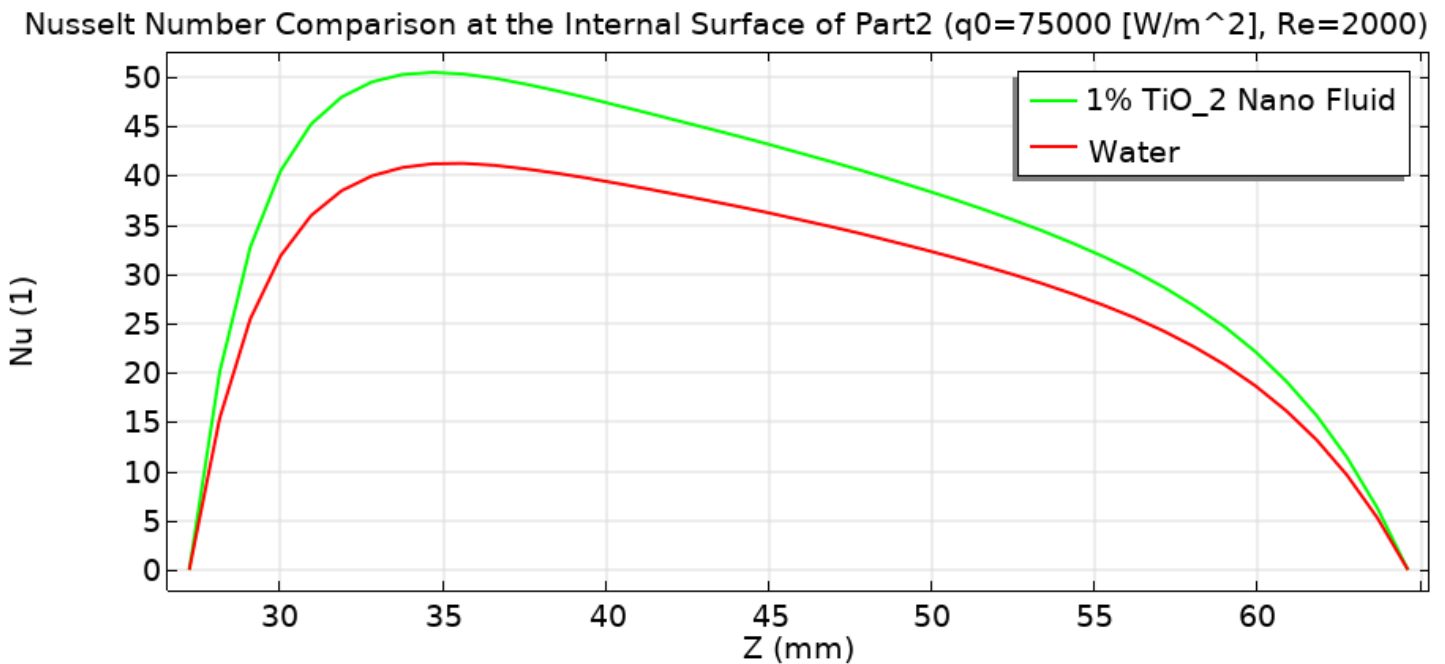


Figure 9 Nusselt Number comparison between Nano Fluid and Water

The nanofluid has a higher Nusselt number as a result of its lower temperature difference observed when compared to water as the working fluid.

4.3. Friction Factor Evaluation

An important parameter to evaluate is the friction factor, which shows the impact of the working fluid properties and the flow path configuration on the pumping power. This factor is defined with the following formula and is directly solved in COMSOL software using the result data expression menu.

$$f = \frac{2 * \Delta P}{\frac{L}{D} * \rho * w^2} \quad (4.1)$$

Where ΔP is the pressure difference between the inlet and the outlet of the pipe, L is the length of the pipe, D is the internal diameter of the pipe and w is the velocity variation inside the pipe in the z-direction. For this case study, the friction factor has been evaluated at the centerline of the pipe. The figure below shows the effect of changing the Reynolds number with water as the working fluid.

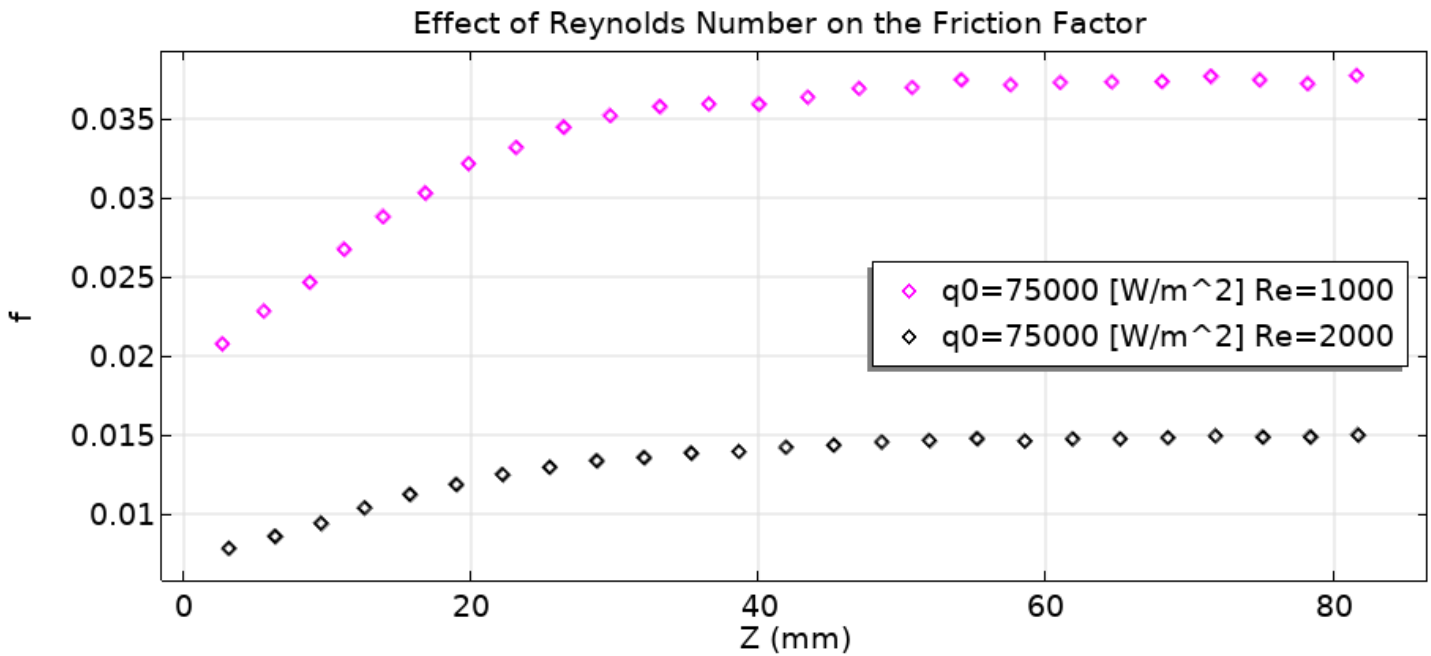


Figure 10 Friction Factor for Water (for radius, $r=0$)

From the plot, it is observed that the effect of friction is more obvious as the flow develops rather than at the initial inlet position through the centerline of the pipe. Thus, it can be said that the flow receives little or no resistance at the initial stage and close to the centerline of the pipe. And this is true when the flow rate is increased by increasing the Reynolds number, in which the friction factor reduces. But this effect is reversed when the flow is studied away from the center line and towards the pipe wall, as shown below for an arbitrary radial location of 3mm away from the pipe centerline.

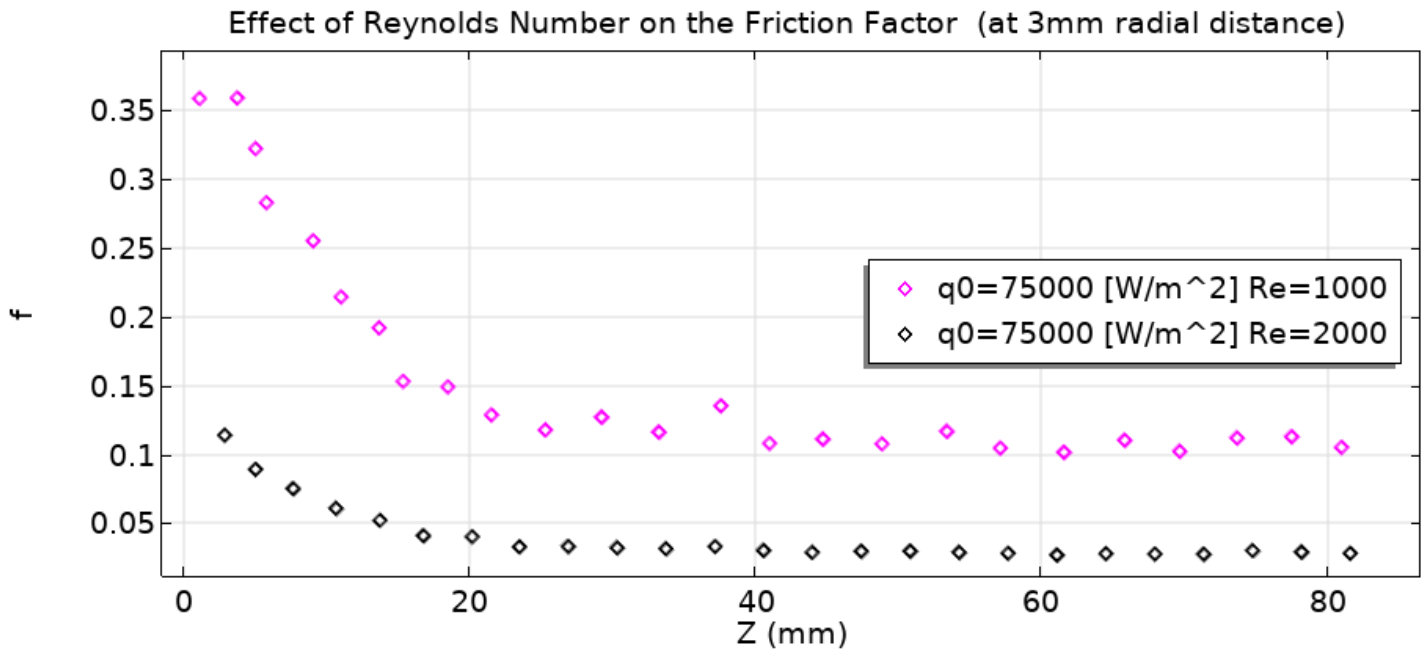


Figure 11 Friction Factor for Water (for radius, $r=3$ mm)

Therefore at the same heat flux and laminar flow condition the friction factor comparison is made for the two working fluids at the pipe centerline (radial distance, $r = 0$), as shown in the figure below:

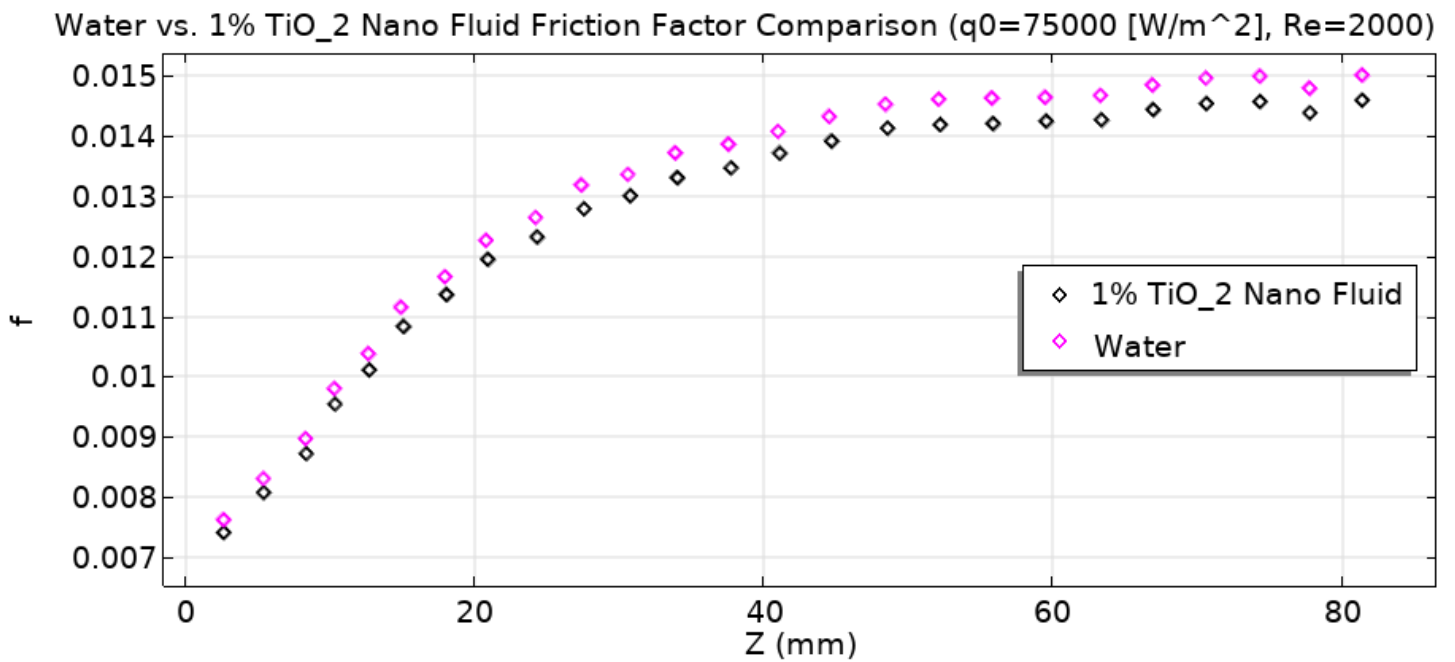


Figure 12 1% TiO₂ Nano Fluid vs. Water Friction Factor

The plot shows that the friction factor for both fluids is very close, but that of the nanofluid is a bit lower because of its slightly higher density, as indicated by equation (4.1) because the pressure difference was slightly lower for the water than for the nanofluid.

4.4. Performance Evaluation for the Working Fluids

The performance evaluation for the two fluids being studied is done using the thermal efficiency ratio. And the thermal efficiency for each case study can be evaluated accurately with the following;

$$\text{Thermal Efficiency} = \frac{Nu_{average} * Z}{f * D}$$

Where, $Nu_{average}$ is the average Nusselt number for any given configuration, and f is the friction factor at each point along the centerline of the pipe in the z-direction, as previously calculated. The following plots were obtained for the heated portion of the pipe:

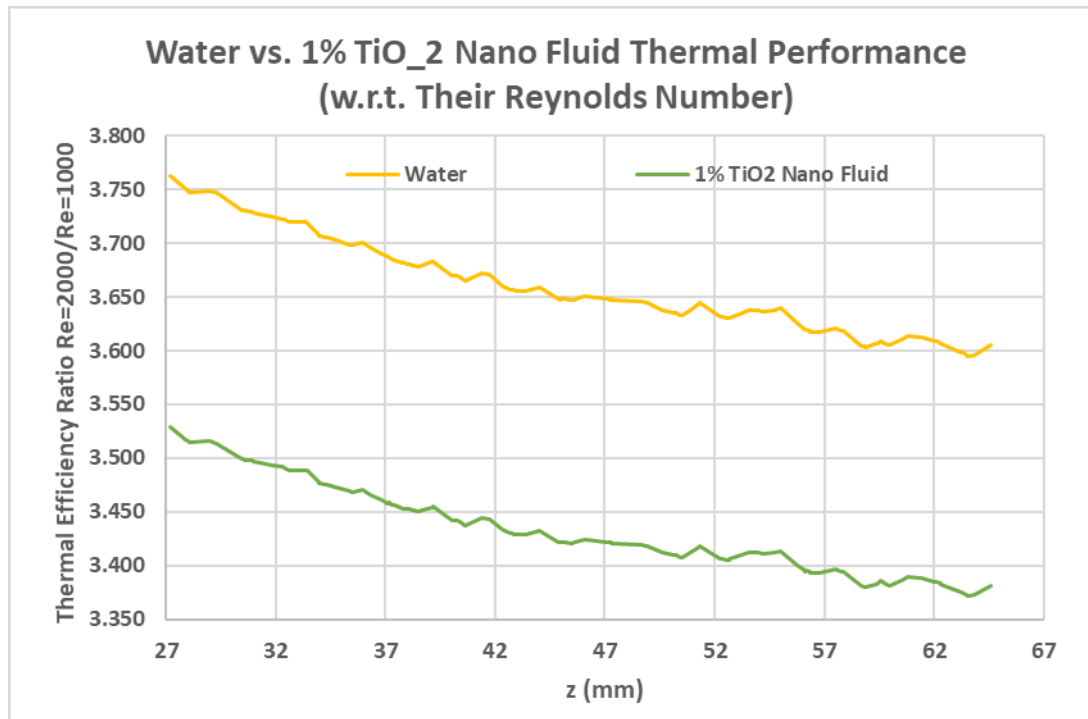


Figure 13 Water vs. Nano Fluid Thermal Performance (w.r.t. Their Reynolds Number)

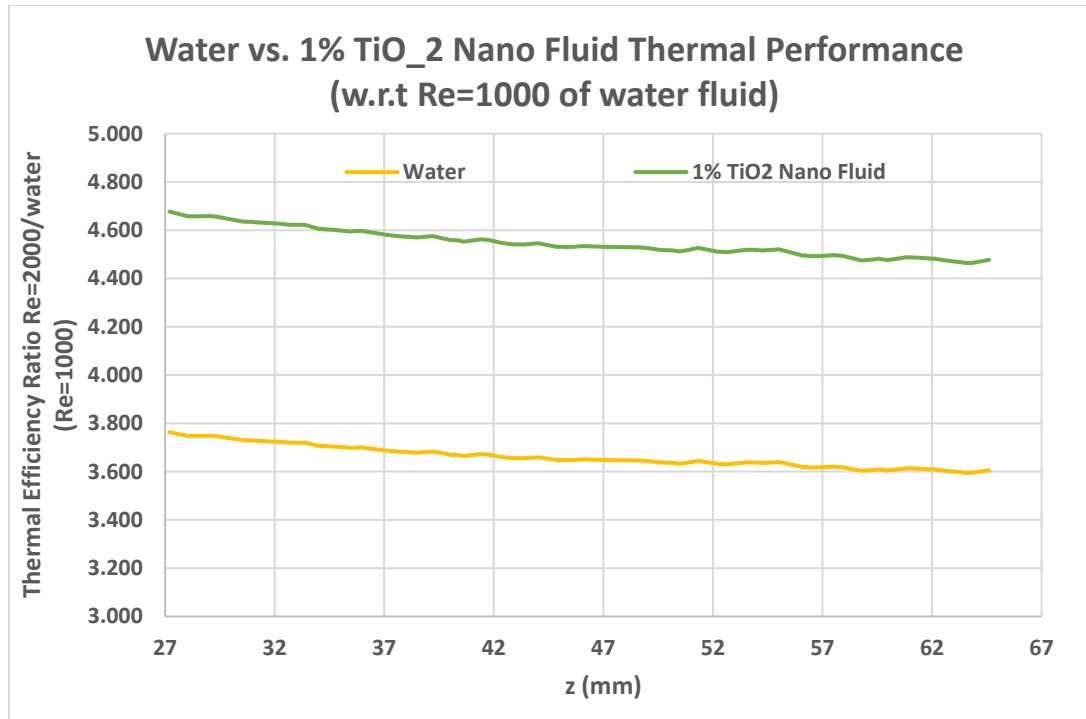


Figure 14 Water vs. Nano Fluid Thermal Performance (w.r.t Re=1000 of water fluid)

The above results show that the thermal effectiveness of changing the Reynolds number from 1000 to 2000 for 1% TiO_2 nanofluid is less when compared to using water as the working fluid. But the nanofluid still has the overall thermal performance, as a result of its higher thermal efficiency ratio when compared against the same water Reynolds number of 1000 (as shown in [Figure 14](#)). This is as a result of the higher Nusselt number and lower friction factor of the nanofluid in comparison with water as previously discussed.

5. Conclusion

The following concluding remarks can be drawn:

- I. The temperature distribution profile is non-linear, which is also a result of the parabolic heat flux input, as well as the convection effect of the cooling fluid flow.
- II. The Nusselt number is independent of the amount of heat flux at the heat part, but it is rather heavily impacted by the fluid condition of the working fluid.
- III. The friction factor difference between the two working fluids doesn't say much on its own. But when used in evaluating the fluid thermal efficiency, it and the Nusselt number does show that the 1% TiO_2 nanoparticle in water base fluid does have a better overall thermal performance than using water alone. But the effectiveness of increasing the Reynolds number is more pronounced in water than in the nanofluid.

6. References

- [1] K. Papazian, Z. Al Hajaj and M. Z. Saghir, "Thermal Performance of a Heated Pipe in the Presence of a Metal Foam and Twisted Tape Inserts," *Fluids*, vol. 5, no. 4, p. 195, 10 2020.
- [2] M. Karami, M. A. Akhavan-Behabadi, M. Raisee Dehkordi and S. Delfani, "Thermo-optical properties of copper oxide nanofluids for direct absorption of solar radiation," *Solar Energy Materials and Solar Cells*, vol. 144, pp. 136-142, 1 2016.
- [3] Z. Alhajaj, A. M. Bayomy and M. Z. Saghir, "A comparative study on best configuration for heat enhancement using nanofluid," *International Journal of Thermofluids*, Vols. 7-8, 11 2020.
- [4] Comsol, "Introduction to COMSOL Multiphysics," 2020.

APPENDIX A

```
clear; clc
%% Variable Initialization
q0=[50000,75000];
Tin=18+273.15; %Temperature at the inlet
t=0.00006; %wall thickness
L2=0.0374; %Length of heated part 2
L1=0.0272; %Length of part 1
k_s=237; %Thermal conductivity of aluminum
n=L2/50; %number of grid point along z-direction
z=(0:n:L2);
zn=z-L2/2;
size=size(z);
col=size(2);
fig1 = figure(1); ax1 = axes; hold(ax1,'on');
%% Loop through the two independent heat flux
for j=1:2
    if j==1
        for i=1:col %Loops through each grid point for q0_1
            q=q0(j)*(1-(4*zn(i)^2/L2^2)); % from equation (1.1)
            T(i)=(q*t/k_s)+Tin; % from equation (3.1)
        end
        plot(ax1,(z+L1)*1000,T-273.15)
    else
        for i=1:col %Loops through each grid point for q0_2
            q=q0(j)*(1-(4*zn(i)^2/L2^2)); % from equation (1.1)
            T(i)=(q*t/k_s)+Tin; % from equation (3.1)
        end
        plot(ax1,(z+L1)*1000,T-273.15)
    end
    %Plot settings
    title('Analytical Solution for Simple Heat Conduction');
    xlabel('Z $(mm)$','Interpreter','latex','fontsize',14);
    ylabel('T $(degC)$','Interpreter','latex','fontsize',14);
    leg1 = legend('$q_0=50000$', '$q_0=75000$');
    set(leg1,'Interpreter','latex');
end
end
```